

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 8: Aircraft Oxygen Equipment (Crew & Passenger)

Syllabus Source: CAE Oxford (pp. 259-280) | Pass Target: >= 90% AviationExam

1. FLIGHT CREW VS PASSENGER OXYGEN REQUIREMENTS

Aviation oxygen systems are divided into two completely segregated architectures due to differing physiological and emergency demands:

System Feature	Flight Crew Oxygen System	Passenger Emergency Oxygen System
Supply Source	High-pressure gaseous oxygen cylinder (1,800 to 2,000 psi) painted green.	Chemical oxygen generators (sodium chlorate candles) mounted in PSUs; or central gaseous cylinder in some widebodies.
Delivery Mode	Diluter-Demand (normal) or 100% Continuous Pressure-Demand (smoke/fire or cabin alt > 39,000 ft).	Continuous flow into a rebreather bag mask.
Deployment	Manual donning. Quick-donning masks stowed beside pilot seats; must be don-able in LESS THAN 5 SECONDS with one hand!	Automatic drop-down when cabin pressure altitude exceeds 14,000 ft; can also be deployed manually from cockpit switch.
Duration	Sized for entire emergency descent down to 10,000 ft plus prolonged cruise at 10,000 ft (typically > 2 hours).	12 to 15 minutes minimum duration (sufficient for emergency descent from FL 410 down to safe breathing altitude < 10,000 ft).
Smoke Protection	Provides positive overpressure inside mask to prevent toxic cockpit smoke/fumes from entering. Built-in smoke goggles.	NOT suitable for smoke/toxic fumes (ambient air enters dilution holes).

[!] EXAM TRAP (AVIATIONEXAM) - Chemical Oxygen Generators (Sodium Chlorate Candles) Hazards

- Chemical Reaction: Sodium Chlorate (NaClO_3) + Iron (Fe) $\rightarrow$ Sodium Chloride (salt) + Oxygen (O_2) + Heat! - The reaction is EXOTHERMIC. Generator core temperatures reach 230°C to 260°C (450°F)! - Activation: Pulling the mask lanyard releases a spring-loaded firing pin that strikes a percussion cap. - CRITICAL: Once initiated, a chemical oxygen generator CANNOT BE SHUT OFF OR EXTINGUISHED until fully consumed (12-15 min)!

2. FLIGHT CREW MASK MODES: NORMAL VS 100% VS EMERGENCY

The flight crew oxygen regulator provides three selectable switch positions:

- **NORMAL (Diluter Demand):** Supplies a variable mix of ambient cabin air and pure oxygen. The regulator automatically increases oxygen concentration as cabin altitude increases, reaching 100% oxygen at approximately 32,000 ft cabin altitude.
- **100% OXYGEN:** Supplies 100% pure oxygen on inhalation demand, bypassing all ambient cabin air. Selected immediately in rapid decompression or contamination.
- **EMERGENCY (Pressure Demand):** Supplies 100% pure oxygen under continuous POSITIVE PRESSURE into the mask. Forces oxygen into the pilot's lungs and forces any smoke or toxic fumes out around mask edges. Mandatory when flying through dense cockpit smoke or above 39,000 ft cabin altitude.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Chemical Oxygen vs Demand Regulator**

1. Passenger Oxygen (Chemical Candle): - Activated by pulling mask lanyard (firing pin strikes chlorate/iron candle). - Produces continuous flow of hot oxygen for 12 to 15 minutes. CANNOT be shut off once started! 2. Flight Crew Regulators: - Normal: Diluter-demand (mixes oxygen with ambient cabin air based on cabin altitude). - 100% / Emergency: Pressure-demand (pure oxygen under positive overpressure to keep toxic smoke out of mask).

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Once initiated by pulling the mask lanyard, can a passenger chemical oxygen generator be turned off by the flight crew?**

- [A]: Yes, by turning off the passenger oxygen switch in the cockpit
- [B]: No, the chemical reaction is exothermic and self-sustaining; it cannot be stopped until all chemical fuel is exhausted (12-15 minutes)
- [C]: Yes, by closing the overhead PSU door
- [D]: Yes, by dumping cabin pressure

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Chemical oxygen generators use an exothermic chemical burning process (sodium chlorate and iron). Once the firing pin is released, the reaction produces oxygen and intense heat (~250°C) continuously until depleted.

Q2: Under EASA CS-25 certification requirements, a flight crew quick-donning oxygen mask must be capable of being donned within how many seconds?

- [A]: 3 seconds
- [B]: 5 seconds
- [C]: 10 seconds
- [D]: 15 seconds

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Regulations mandate that a pilot must be able to don and latch a quick-donning oxygen mask using only one hand within 5 seconds, while continuing to fly and communicate.

Q3: What is the primary purpose of the 'EMERGENCY' setting on a flight crew oxygen regulator?

- [A]: To save oxygen supply during long descents
- [B]: To provide 100% oxygen under continuous positive pressure, preventing toxic fumes and smoke from entering the mask
- [C]: To inflate the passenger cabin
- [D]: To heat the oxygen gas

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Emergency mode delivers continuous positive pressure, which prevents toxic smoke, carbon monoxide, or irritant gases from seeping past the mask facial seal into the pilot's respiratory tract.